
Nano-ESMC: A Minimal, Hill-Climbable ESMC Training Reproduction

Lumin Science
<https://luminscience.io>

Abstract

Nano-ESMC is a compact, end-to-end reconstruction of ESMC-style protein language-model training. It connects a public 665,970,495-protein, evaluation-decontaminated corpus to a fixed four-GPU training contract and frozen biological evaluation. Full long-range contact precision at L (P@L) on 20,775 RCSB Protein Data Bank chains is the primary model-selection axis; held-out masked-language-model loss and the four-task P-CORE representation panel are supporting measurements. Under the current one-hour four-L40S contract, the original ESMC-300M-class baseline reaches 0.0925 P@L and the first retained AutoResearch candidate reaches 0.0960. Released ESMC-300M, 600M, and 6B checkpoints establish the external scale ladder for P-CORE, while a source-qualified P-CORE Next alpha campaign records which new tasks discriminate scale and which fail qualification. This report presents the data pipeline once, the evaluation contract once, and all baseline evidence in one section.

1 Scope and fixed contract

The goal is to reproduce the essential ESMC sequence-model training path in a form that is small enough to inspect and strict enough to hill climb [Candido et al., 2026]. A candidate may change model internals, optimization, batching, kernels, or schedules, but it cannot change the data, compute budget, evaluation examples, or metric after observing a result.

Object	Current contract
Model class	Sequence-only ESMC-300M encoder trained from scratch; no structure-conditioned input.
Data	Immutable public UniRef90, MGnify, and IMG/IMG release with exact and homology decontamination against the complete protected evaluation union.
Compute	Four NVIDIA L40S GPUs and 3,600 seconds of synchronized training-loop time.
Primary selection	Full long-range contact P@L on the frozen 20,775-chain population.
Supporting axes	Held-out sequence-mean MLM loss and the four-task P-CORE frozen-representation panel.
Promotion	Strict P@L improvement at matched compute, no unexplained trusted-task regression, then clean rebuilds and repeated seeds before a release claim.

Table 1: The public Nano-ESMC experiment contract.

The current repository supports Stage-1 training at context length 512. Longer-context Stage 2, larger local models, and week-scale compute projections are outside the present claim. Historical experiments remain evidence about scaling, not alternate active contracts.

2 Data: one corpus and one processing pipeline

2.1 Sources and the main gap to ESMC

ESMC reports training pools from UniRef, MGnify, and a July 2023 Joint Genome Institute (JGI) snapshot [Candido et al., 2026, Suzek et al., 2015, Richardson et al., 2023]. Nano-ESMC reconstructs the same source roles with UniRef90 2023_02, MGnify 2023_02, and the IMG/JGI-labelled portion of the public OMG corpus [Cornman et al., 2024]. OMG/IMG is a surrogate for the JGI role, not the authors’ exact snapshot.

Source role	ESMC 70%-identity clusters	Nano-ESMC representatives	Nano / ESMC
UniRef	approximately 83M	92,230,941	approximately 111%
MGnify	approximately 372M	348,135,082	approximately 94%
JGI role / OMG-IMG surrogate	approximately 2B	324,923,979	approximately 16%
Total	approximately 2.455B	765,290,002	approximately 31%

Table 2: Stage-aligned comparison with the cluster counts reported by ESMC. The roughly 1.68B-representative JGI-role gap dominates the difference.

2.2 Processing and release accounting

The complete path is fixed in seven steps:

1. pin every upstream object and its digest;
2. normalize sequences and assign OMG accessions to one source role;
3. reject sequences shorter than 60 amino acids or with more than 20% ambiguous residues;
4. collapse exact sequences while retaining source membership;
5. cluster each source view independently at 70% identity and 80% shorter-sequence coverage with MMseqs2 Linclust [Steinegger and Soeding, 2018];
6. remove exact matches and homologs to the protected evaluation union; and
7. assign cross-source ownership, select validation rows, write deterministic Parquet shards, and independently verify every digest.

Source	Original records	Quality eligible	Distinct in source	70% representatives	After eval. decontam.	Final training
UniRef90	170,669,877	165,884,293	165,884,293	92,230,941	74,181,938	74,175,974
MGnify	729,215,663	611,817,034	611,788,129	348,135,082	330,444,705	328,949,335
OMG/IMG	3,280,269,924	2,085,746,535	963,673,186	324,923,979	282,202,559	262,845,186
Total source views	4,180,155,464	2,863,447,862	1,741,345,608	765,290,002	686,829,202	665,970,495

Table 3: The single corpus funnel. Source-view distinct counts can overlap because one exact sequence may carry more than one source membership. Cross-source ownership is resolved before the final training count.

The protected union contains 317,000 unique evaluation sequences, including all probe-fit, validation, gallery, test, blocked-candidate, and contact rows known at release-build time. It includes the 20 contact-probe chains and all 20,775 contact-evaluation chains from the frozen 2024-02-28 RCSB PDB snapshot. The pipeline removes 21,653 exact matches and 78,439,147 additional homologs at least 30% identity, at least 80% bidirectional coverage, and $E\text{-value} \leq 10^{-3}$ using MMseqs2 [Steinegger and Söding, 2017]. After decontamination, 20,844,431 non-owner source memberships, 1,988 out-of-range records, and 12,288 validation rows are excluded to reach the final training release.

The immutable public artifact contains 565 training shards, 665,970,495 proteins, and 151,304,238,405 residues. Each source contributes one additional 4,096-protein validation shard. The complete train/validation release occupies 109,661,312,410 compressed bytes. These values define the supported corpus. The earlier 9M-protein materialization appears only in the historical baseline table in Section 4.

3 Evaluation

3.1 Three reported axes

Axis	Role	Definition
Held-out MLM	Diagnostic	Sequence-mean negative log-likelihood and perplexity on 12,288 hash-disjoint validation representatives.
P-CORE	Representation panel	Four frozen-encoder tasks spanning evolution, local structure, localization, and fitness.
Long-range contact P@L	Primary selection	Mean top- L precision over the frozen 20,775-chain population using all-layer/all-head attention features.

Table 4: Evaluation axes and their decision roles. A lower MLM loss does not override a contact P@L regression.

3.2 Contact P@L

All contact structures and labels come from the 2024-02-28 RCSB Protein Data Bank snapshot [Berman et al., 2000]. Following the ESM contact protocol and the ESMC long-range definition [Rao et al., 2021, Candido et al., 2026], all attention heads and layers are symmetrized and fed to a logistic probe. Sixteen structures fit candidate probes, four select regularization, the selected probe is refit on all 20, and 20,775 chains supply the final score. A true contact has $C\beta$ distance below $8\text{\AA}$ ($C\alpha$ for glycine), long range means sequence separation at least 24, and the per-chain metric is precision among the top L predicted pairs.

The snapshot yields 216,478 source structures and 781,077 extracted protein chains before filtering. Chains are reduced at 40% sequence identity and 80% shorter-sequence coverage, truncated to 510 residues, and required to contain at least L eligible true contacts. The final population contains 20,775 chains representing 20,758 unique sequences. ESMC reports the same snapshot date, population size, and construction rules, but does not publish its ordered chain manifest or raw-file digests. The two evaluations are protocol-matched, not proven item-for-item identical.

3.3 P-CORE

The encoder remains frozen. Protein tasks mean-pool final-layer residue vectors; secondary structure uses residue vectors directly. A standardizer and a fixed low-capacity logistic or ridge probe are fit on probe-training data, regularization is selected on validation, and the selected probe is applied once to the untouched test set. Bootstrap intervals resample frozen test predictions by biological unit and do not refit the encoder or probe. The task sources are DeepSF/TAPE for remote homology [Hou et al., 2018, Rao et al., 2019], NetSurfP-2.0/CB513 for secondary structure [Klausen et al., 2019, Cuff and Barton, 1999], DeepLoc2 for localization [Thummuluri et al., 2022], and FLIP2 for fitness [Didi et al., 2026].

Task	Fit / validation / test	Frozen readout	Primary metric	Unit
Remote homology	12,312 / 736 / 718 proteins	mean + multinomial logistic	balanced accuracy	family group
Secondary structure	8,678 / 2,170 / 513 proteins	residue + logistic	residue macro-F1	protein
DeepLoc2	28,303 proteins in five rotating folds	mean + multilabel logistic	macro average precision	fold / protein
FLIP2 Hydrophobic Core	9,974 / 2,493 / 12,468 variants	mean + ridge	Spearman correlation	variant group

Table 5: The four trusted P-CORE tasks. Each raw metric is null-normalized; the score is 100 times their geometric mean. A task at or below its null invalidates the aggregate.

Enzyme Commission and the reconstructed Human PPI benchmark remain visible diagnostics, but neither can affect P-CORE or model selection. Their released model behavior is reported once in Section 4.

4 Baseline and qualification evidence

4.1 Current one-hour AutoResearch baseline

Model	P@L	Delta	Train loss	Valid. loss	Steps	Tokens (M)	Params. (M)	Peak GB	Train h
Original ESMC	0.0925	–	2.723	2.699	5,883	358	333	35	1
AutoResearch-Codex-Round1	0.0960	+0.0036	2.720	2.704	5,682	346	333	37	1

Table 6: Compute-matched Stage-1 runs on four L40S GPUs. P@L is the keep or reject metric; losses are required diagnostics.

The retained candidate adds learned residual/input routing, parameter-free transformer RMSNorm, depth-scaled residual initialization, and a linear learning-rate cooldown over the final 20% of training. Its improvement is a single fixed-checkpoint result, not a multi-seed release claim.

4.2 Released ESMC checkpoints and quarantined diagnostics

Metric	ESMC-300M	ESMC-600M	ESMC-6B	Role
Remote homology balanced accuracy	0.1159	0.1152	0.1186	P-CORE
Secondary-structure macro-F1	0.8315	0.8407	0.8780	P-CORE
DeepLoc2 macro average precision	0.6442	0.6568	0.6942	P-CORE
FLIP2 Hydro Spearman	0.4132	0.4276	0.4616	P-CORE
P-CORE	37.4847	38.1553	40.6011	reference panel
Enzyme Commission macro average precision	0.7174	0.7096	0.0237	quarantined
Human PPI average precision	0.8155	0.8026	0.8345	quarantined
ESMC paper contact P@L-LR	0.552 ± 0.002	0.589 ± 0.002	0.725 ± 0.002	published
Nano-ESMC full 20,775-chain P@L	0.5387	0.5803	pending	local protocol

Table 7: One consolidated released-model reference table. P-CORE values are local recomputations with the frozen probes. Contact paper values come from ESMC; local contact values use the Nano-ESMC chain manifest.

P-CORE is ordered across the three released scales, although remote homology is nearly flat. EC collapses at 6B despite clean split and embedding-integrity checks. Human PPI has only 237 test pairs, is non-monotone across scale, and was nearly matched by a severely undertrained local model. These behaviors justify quarantine rather than post hoc repair. The full 6B local contact evaluation remains pending and is not replaced by the completed 1,024-chain diagnostic.

4.3 Historical training-duration check

Historical run	Steps	Model tokens	P-CORE	Contact P@L	Decision
Local ESMC-300M, 4 h	20,971	1.290B	25.4609	0.1013	baseline
Local ESMC-300M, 16 h	84,000	5.166B	30.6609	0.1636	improved, not promoted
Released ESMC-300M	–	–	37.4847	0.5387	external reference

Table 8: Historical four-A100 duration experiment. Both local runs used the superseded 9M-protein materialization, so they are retained as scaling evidence and are not baselines for the current public corpus.

All four trusted raw tasks improved from four to sixteen hours. Contact P@L increased by 0.0623, with a fixed-checkpoint paired 95% interval of [0.0614, 0.0632]. The runs used different seeds, so that interval does not isolate training duration from initialization. The sixteen-hour checkpoint remained below released ESMC-300M on P-CORE and contact and was not promoted.

4.4 P-CORE Next alpha qualification

The expansion campaign evaluated identical manifests with released ESMC-300M, 600M, and 6B checkpoints and 10,000 paired biological-unit bootstrap resamples. Candidate sources are CATH 4.4 [Waman et al., 2025], PRING [Zheng et al., 2025], FLIP2 [Didi et al., 2026], MegaScale [Tsuboyama et al., 2023], CAID3 [Mehdiabadi et al., 2026], and CAFA5 [De Paolis Kaluza et al., 2026]. It is a qualification profile, not a new aggregate.

Candidate / metric	ESMC-300M	ESMC-600M	ESMC-6B	Decision
CATH 4.4 macro-H mAP	0.1170	0.1067	0.1153	diagnostic; not scale-monotone
PRING Human AUPRC	0.7035	0.7001	0.7113	provisional; intervals failed centering
FLIP2 hierarchical score	0.7823	0.7581	0.7805	diagnostic; limited landscapes
MegaScale median-parent Spearman	0.6835	0.7267	0.7207	qualification candidate
CAID3 macro-protein AP	0.7292	0.7341	0.7340	temporal diagnostic
CAFA5 molecular function	blocked	blocked	blocked	110 hard proteins < 500 gate

Table 9: P-CORE Next alpha results. Missing tasks are not imputed and no aggregate or released-model ranking is issued.

No runnable candidate is strictly monotone across all three scales. MegaScale is the only new task with a clear paired 6B-over-300M difference: $+0.0372$ with a 95% interval of $[0.0064, 0.1171]$. PRING has only one negative realization and non-centered intervals. CAFA5 failed its frozen minimum-population gate. CATH, FLIP2 shift, and CAID3 remain useful diagnostics but do not yet qualify as promotion gates.

5 Limitations and next work

The public corpus reproduces ESMC’s source roles and processing logic, but its OMG/IMG surrogate is much smaller than the reported JGI cluster pool and cannot reproduce ESMC’s unavailable cluster-member sampling. The released contact protocol matches the ESMC description, snapshot date, and population size, but the unpublished Biohub chain manifest prevents an item-for-item identity claim. The current one-hour AutoResearch improvement is single-run evidence. The historical duration runs use a superseded corpus. The full released ESMC-6B contact value is still pending. P-CORE Next has not qualified a successor aggregate.

The immediate work is therefore:

1. complete the full 6B contact measurement;
2. repeat the one-hour incumbent and strongest candidate across seeds;
3. requalify the four current P-CORE tasks under the same gates applied to new candidates;
4. complete MegaScale confound, weak-model, and probe-stability controls;
5. repair PRING with a preregistered uncertainty estimator and ten negative realizations; and
6. shadow any P-CORE successor beside the current four-task panel before changing model-selection policy.

Nano-ESMC should be read as a reproducible research system and a measured hill-climbing baseline. It does not yet claim an exact ESMC data reproduction, a reproduction-quality checkpoint, or a replacement P-CORE suite.

References

- Helen M. Berman, John Westbrook, Zukang Feng, Gary Gilliland, T. N. Bhat, Helge Weissig, Ilya N. Shindyalov, and Philip E. Bourne. The protein data bank. *Nucleic Acids Research*, 28(1):235–242, 2000. doi: 10.1093/nar/28.1.235.
- Salvatore Candido, Thomas Hayes, Roshan Rao, et al. Language modeling materializes a world model of protein biology. *bioRxiv*, 2026. doi: 10.64898/2026.06.03.729735.
- Andre Cornman, Jacob West-Roberts, Antonio Pedro Camargo, Simon Roux, Martin Beracochea, Milot Mirdita, Sergey Ovchinnikov, and Yunha Hwang. The OMG dataset: An open metagenomic corpus for mixed-modality genomic language modeling. *bioRxiv*, 2024. doi: 10.1101/2024.08.14.607850.

- James A. Cuff and Geoffrey J. Barton. Evaluation and improvement of multiple sequence methods for protein secondary structure prediction. *Proteins: Structure, Function, and Genetics*, 34(4):508–519, 1999.
- M. Clara De Paolis Kaluza, Rashika Ramola, Parnal Joshi, Damiano Piovesan, Walter Reade, Sandra Orchard, Maria J. Martin, et al. Advances in protein function prediction from the fifth CAFA challenge. *bioRxiv*, 2026. doi: 10.64898/2026.04.27.716980.
- Kieran Didi, Sarah Alamdari, Alex X. Lu, Bruce Wittmann, Kadina E. Johnston, Ava P. Amini, Ali Madani, Maya Czeneszew, Christian Dallago, and Kevin K. Yang. FLIP2: Expanding protein fitness landscape benchmarks for real-world machine learning applications. *bioRxiv*, 2026. doi: 10.64898/2026.02.23.707496.
- Jie Hou, Badri Adhikari, and Jianlin Cheng. DeepSF: deep convolutional neural network for mapping protein sequences to folds. *Bioinformatics*, 34(8):1295–1303, 2018. doi: 10.1093/bioinformatics/btx780.
- Michael Schantz Klausen, Martin Closter Jespersen, Henrik Nielsen, Kamilla Kjaergaard Jensen, Vanessa Isabell Jurtz, Casper Kaae Soenderby, Morten Otto Alexander Sommer, Ole Winther, Morten Nielsen, Bent Petersen, and Paolo Marcatili. NetSurfP-2.0: Improved prediction of protein structural features by integrated deep learning. *Proteins: Structure, Function, and Bioinformatics*, 87(6):520–527, 2019. doi: 10.1002/prot.25674.
- Mahta Mehdiabadi, Alessio Del Conte, Maria Victoria Nugnes, Maria Cristina Aspromonte, Silvio C. E. Tosatto, and Damiano Piovesan. Critical assessment of protein intrinsic disorder round 3—predicting disorder in the era of protein language models. *Proteins: Structure, Function, and Bioinformatics*, 94(1):414–424, 2026. doi: 10.1002/prot.70045.
- Roshan Rao, Nicholas Bhattacharya, Neil Thomas, et al. Evaluating protein transfer learning with TAPE. In *Advances in Neural Information Processing Systems*, 2019.
- Roshan Rao, Joshua Meier, Tom Sercu, Sergey Ovchinnikov, and Alexander Rives. Transformer protein language models are unsupervised structure learners. In *International Conference on Learning Representations*, 2021.
- Lorna Richardson, Ben Allen, Giuseppe Baldi, et al. MGnify: the microbiome sequence data analysis resource in 2023. *Nucleic Acids Research*, 51(D1):D753–D759, 2023. doi: 10.1093/nar/gkac1080.
- Martin Steinegger and Johannes Söding. MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35:1026–1028, 2017. doi: 10.1038/nbt.3988.
- Martin Steinegger and Johannes Soeding. Clustering huge protein sequence sets in linear time. *Nature Communications*, 9: 2542, 2018. doi: 10.1038/s41467-018-04964-5.
- Baris E. Suzek, Yuqi Wang, Hongzhan Huang, Peter B. McGarvey, Cathy H. Wu, and the UniProt Consortium. UniRef clusters: a comprehensive and scalable alternative for improving sequence similarity searches. *Bioinformatics*, 31(6):926–932, 2015. doi: 10.1093/bioinformatics/btu739.
- Vineet Thummuluri, Jose Juan Almagro Armenteros, Alexander Rosenberg Johansen, Henrik Nielsen, and Ole Winther. DeepLoc 2.0: multi-label subcellular localization prediction using protein language models. *Nucleic Acids Research*, 50 (W1):W228–W234, 2022. doi: 10.1093/nar/gkac278.
- Kotaro Tsuboyama, Justas Dauparas, Jonathan Chen, Elodie Laine, Yasser Mohseni Behbahani, Jonathan J. Weinstein, Niall M. Mangan, Sergey Ovchinnikov, Gabriel J. Rocklin, et al. Mega-scale experimental analysis of protein folding stability in biology and design. *Nature*, 620:434–444, 2023. doi: 10.1038/s41586-023-06328-6.
- Vaishali P. Waman, Nicola Bordin, Andy Lau, Shaun Kandathil, Jude Wells, David Miller, Sameer Velankar, David T. Jones, Ian Sillitoe, and Christine Orengo. CATH v4.4: major expansion of CATH by experimental and predicted structural data. *Nucleic Acids Research*, 53(D1):D348–D355, 2025. doi: 10.1093/nar/gkac1087.
- Xinzhe Zheng, Hao Du, Fanding Xu, Jinzhe Li, Zhiyuan Liu, Wenkang Wang, Tao Chen, Wanli Ouyang, Stan Z. Li, Yan Lu, Nanqing Dong, and Yang Zhang. PRING: Rethinking protein-protein interaction prediction from pairs to graphs. In *Advances in Neural Information Processing Systems*, volume 38, 2025.